# Helix Grid Systems

Grid-edge orchestration software for electric utilities

Series B Investor Presentation | Confidential

# The Problem

Utilities must balance four times more distributed energy resources with dispatch tools built for one-way grids.

Grid operators lose an estimated $9B annually to curtailment and manual balancing.

# Product & Traction

HelixOS is in production at 38 utilities and generally available since 2023.

$14.2M ARR with 118% YoY growth and 78% gross margin.

Largest customer represents 9% of revenue; net revenue retention of 131%.

Processes 2.1 billion telemetry events per day across 1.4 million grid devices.

# Platform Architecture

Cloud-native microservices on Kubernetes, deployed multi-region on AWS and Azure with active-active failover.

Public REST and gRPC APIs with SDKs for utility integrations.

Infrastructure as code with Terraform; observability with Datadog and OpenTelemetry.

Autoscaling event pipeline load-tested to 10x current peak volume.

| Layer | Technology |
|---|---|
| Ingestion | Kafka event streaming |
| Forecasting | Proprietary forecasting models built in-house |
| Data | PostgreSQL and TimescaleDB |

# Engineering & Delivery

CTO Dr. Maya Chen led grid software engineering at Siemens for nine years.

46 full-time in-house engineers across Denver and Toronto.

CI/CD with GitHub Actions; 84% automated test coverage; weekly releases; mandatory code reviews.

99.95% uptime SLA over the trailing 24 months.

Incident response with on-call rotation and blameless post-mortems.

Architecture documentation, runbooks and decision records maintained for every service.

Engineering leadership: CTO, VP Engineering, Director of SRE

# Security & Compliance

SOC 2 Type II attested annually and ISO 27001 certified.

Annual third-party penetration testing and a public bug bounty program.

Data encrypted with AES-256 at rest and TLS 1.3 in transit; SSO, MFA and role-based access control.

Disaster recovery tested quarterly with 15-minute RPO and 1-hour RTO.

# Intellectual Property

7 issued U.S. patents and 5 PCT applications covering forecasting and dispatch methods.

All patents are assigned to Helix Grid Systems, Inc.

All employees and contractors have signed invention assignment agreements.

Proprietary dataset of 11 billion labeled grid telemetry records.

Open-source license scanning with FOSSA in CI; no copyleft components in distributed code.

Independent IP and code audit completed in 2025.

# Market & Competition

Embedded in utility control-room workflows as the system of record for DER dispatch.

High switching costs from integrations with SCADA and ADMS platforms.

# The Round

Raising $40M Series B to expand into European utilities.

Raised $31M to date from climate and infrastructure investors.

Use of funds: EU expansion, FedRAMP authorization planned for 2027, and grid-forecasting R&D.